

Application of remote sensing and GIS in monitoring water quality parameters: A comprehensive review of key parameters

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Abstract - This review comprehensively outlines key considerations for assessing water quality using remote sensing satellite data. It highlights the significance of critical water quality metrics such as chlorophyll-a (chl-a), turbidity, temperature, pH, dissolved oxygen, nitrogen, and phosphorus in remote sensing-based analysis. The paper explores the optical characteristics of these metrics relevant to remote sensing applications by synthesizing insights from over 17 research papers and reviews, this paper serves as a valuable resource for researchers, aiding in the design and implementation of remote sensing-based water quality assessments. The summarized characteristics contribute to advancing water quality assessment and management through remote sensing. This review helps remote sensing obtain essential water quality information for environmental monitoring, resource management, and policy-making. It also highlights the potential of these metrics in addressing global water quality concerns through remote sensing-based studies. Manual water quality testing is expensive and time-consuming so water quality detection through remote sensing helps to save both.

Index Terms - Remote sensing, Spectral reflectance, Inversion model, Satellite data, Correlation analysis, Band ratio, Water Quality, Monitoring, and Concentration estimation.

I. INTRODUCTION

Monitoring and managing aquatic ecosystems, managing water resources, and determining the uses of water based on its quality all require the assessment of water quality. To gather, integrate, analyze, and disseminate data and information for operational applications and commercialization objectives, remote sensing and GIS technologies provide a cost-effective solution [15]. Traditional techniques of water quality monitoring sometimes entail time-consuming and laborious field sampling, which may yield irregular and localized data [5]. By providing synoptic and recurring observations of significant water bodies at diverse spatial and temporal dimensions, remote sensing satellites, in contrast, offer a distinct advantage. This capacity enables an in-depth understanding of the dynamics of water quality, enabling effective monitoring and management techniques.

Despite the inherent trade-off between precision and estimation, remote sensing-based systems for assessing water quality provide many advantages over more conventional approaches. The capability to cover enormous areas, including isolated and harsh areas, is a fundamental asset. This is especially important for large-scale water resource management and environmental monitoring. On a regional or even global scale, remote sensing satellites can offer useful information on water quality indicators such as chlorophyll-a, turbidity, TSS, Turbidity, Temperature & pH.

Furthermore, because remote sensing methods offer temporal data, it is possible to detect dynamic changes in water quality over time. Understanding seasonal fluctuations, seeing long-term patterns, and locating episodic events like algae blooms or pollution issues are all made possible by this temporal dimension. For successful management of water quality, a continual monitoring capability is ensured by the repeated nature of satellite observations. Even while remote sensing-based assessment may only yield approximations rather than exact data, it has a distinct benefit in terms of efficiency. Traditional field-based techniques are frequently quite labor-intensive and logistically challenging, rendering them unsuitable for large-scale studies. In contrast, remote sensing data can be acquired more affordably and quickly, making it possible to track temporal fluctuations and monitor water bodies frequently. In this paper, we will gain an insight into the reasons why so many researchers focus in particular on a river, lake, or other body of water in a certain area.

We will examine the key factors that researchers should consider when evaluating water quality using remote sensing satellite imagery in the following sections. Each metric will be discussed in a standardized manner, with an introduction that emphasizes its importance for determining water quality. After that, we will look at the many methods researchers have used to estimate the parameter using remote sensing data. Additionally, we will outline the limitations and future potential of our inferences based on the evaluation of numerous research publications and review articles. Finally, we will suggest possible solutions and improvements in remote sensing methods to overcome the cited constraints. Throughout the review, we will emphasize the importance of understanding the optical properties and behavior of each parameter, as well as their relationship with water quality.

II. CHLOROPHYLL-A

(a) Introduction

Photosynthesis depends on the pigment chlorophyll-a, which is present in plants, algae, and cyanobacteria. It reflects green light and gives plants their characteristic green hue by absorbing light in the red and blue portions of the electromagnetic spectrum. Due to its optical properties, chlorophyll-a is the primary focus in the majority of research papers. It is found in all photosynthesizing organisms and is the most abundant chlorophyll. Chlorophyll-a is used as a function for total algal biomass in aquatic environments. Additionally, not all algae are harmful; harmful blooms contain particular species [1]. It is well known that it not only endangers aquatic life but also poses a risk to human health, wildlife, and livestock [3]. Traditionally it is analyzed by taking samples and testing in the laboratory,

which is an inefficient way. Since chl-a is optically reactive, a new method for retrieving its concentration has been developed in response to the development of remote sensing technologies.

(b) Methods

The chlorophyll-a retrieving technique is broadly divided into 3 categories, Empirical, analytical, and Machine Learning models. Due to its limitation of non-generalizability across large spatial and temporal scales where variations in atmospheric conditions, the empirical model is further divided into semi-empirical models. These models involve fitting a standard linear regression between spectral band/band ratio values and temporally coincident in situ water quality measurements. Semi-empirical models that use multi-band index values and have some basis in the physical properties of the target constituent do not include any inverse modeling of the inherent optical properties of a particular waterbody. There are several semi-empirical markers, including the normalized difference suspended sediment index, the maximum chlorophyll index [2], the floating algal index, and others. Semi-empirical models are more generalizable than strictly empirical techniques. While Analytical models involve the development of an inverse model with some light properties as the complex parameter, hence it is a further evolved method that is semi-analytical which uses in-situ data to parameterize the equations. Machine learning models originated from empirical models, However, they differ from empirical models in that they can handle complex non-linear interactions [1].

In semi-empirical we have the Band ratio model, the Three-band model, and for the semi-analytical model we have the order differential model, [4] It uses reflectance of bands B4 (red) and B5 (near-infrared) from the Sentinel-2 satellite to determine the content of chlorophyll a (chl-a), the 3BSI includes following formulation

$$3BSI = [R^{-1}(\lambda_1) - R^{-1}(\lambda_2)] \cdot R(\lambda_3) \quad (1)$$

$$Chl-a(\mu gL^{-1}) = 279.95(3BSI) + 38.06 \quad (2)$$

Where, $R(\lambda_i)$:

$i = 1-3$ are the reflectance of the band (4,5,6) [6]

$\lambda_1 \in [660, 690] \text{ nm}$

band-4(water shows maximum absorption in this)

$\lambda_2 \in [700, 730] \text{ nm}$

band-5 (minimum sensitive)

$\lambda_3 \in [740, 760] \text{ nm}$

band-6 (absorption close to clean water)

Likewise, there is the SNAP (Sentinel Application Platform) toolkit which uses C2RCC atmospheric correction and generates product “conc chlorophyll-a” through which we can retrieve chl-a (gm-3), [5]. In Machine learning or Artificial intelligence models, we have many models which include, SVMRM (Support vector regression model), Artificial Neural Network (ANN), curve fitting, backpropagation (BP) neural network, and radialbasis function (RBF) neural network [9]. In [3] researchers have developed a new kind of neural network called a Multitasked Neural Network, whose primary task is to retrieve chlorophyll-a level, and the secondary task is to classify the unlabelled data to its acquisition month, which helps to fillout missing value later. This resulted in an efficient Neural network, for this researcher used sentinel-2 level-1 images. thetechnique involves training a multitasking network with two branches: Month categorization and Chl-a regression are the two key tasks. By simultaneously training the feature extractorfor both tasks, the network learns beneficial features for both, reinforcing their performance. This approach leverages the use of unlabeled data to avoid overfitting and improvegeneralization capabilities [3].

(c) Summary and Limitations

To summarize this, for estimating chl-a we can use multispectral images of remote sensing satellites like Sentinel-2, LandSAT-8, MERIS, MODIS, RapidEye, etc., and analyze this using empirical, semiempirical analytical, and machine learning models, among which semi-empirical are better than semi-analytical, and ML base model outperform than other twomodels, in ML based model SVM BP is the best, and also AI base Multitask Neural network showed a better performance than SVP for large data size, also we have sentinel toolkitto estimate chl-a product. For atmospheric correction, we have polymer and C2RCC (Case 2 Regional Coast Color).

During the evaluation of chlorophyll-a levels, it is importantto address a potential limitation in the current mode, SNAPis more useful if water bodies are large in area, and empirical models are more sensitive towards atmospheric error that is clouds, etc. solely estimating chlorophyll-a may lead to amisleading water quality report, as the green color associated with chlorophyll-a can also come from non-harmful aquatic plants like lotus or water surface vegetation. To differentiate between these vegetation types and harmful algal blooms, a possible solution is to study the patterns of harmful and non-harmful algae blooms and develop a classification system. Thiscomprehensive approach would involve incorporating multiple optical parameters and utilizing classification algorithms to generate accurate water quality reports based on chlorophyll-aestimations.

III. TURBIDITY

(a) Introduction

The presence of suspended particles or solids within a fluid is the main cause of its turbidity, which describes the degree of cloudiness or haziness that it exhibits. When talking about water, turbidity takes on the role of a metric that measures the amount of transparency or clarity present in the water column. The presence and quantity of particulate matter, which can affect numerous ecological and functional characteristics, is a key factor in determining the overall health and quality of aquatic systems. Turbidity precisely can help you gain a thorough grasp of aquatic ecosystems and the dynamic interactions they have with the larger ecosystem. It indicates the amount of particulate matter, such as sediment, silt, clay, organic matter, and microorganisms, that is suspended in the water. Phytoplankton, suspended sediment, organic contaminants, and dissolved chemicals can all have an impact on water quality. The depth to which sunlight may enter the water will be reduced in turbid water. The amount of light that is dispersed or obstructed by suspended particles in water is referred to as turbidity. A crucial evaluation of water quality is the measurement of turbidity.

(b) Methods

To build remote sensing links with turbidity, satellites are routinely used. Notable satellite platforms used in this endeavor include Landsat 5, Landsat 8, and Sentinel-2. The selection of bands located in the red and near-infrared (NIR) areas of the electromagnetic spectrum is where the majority of the turbidity research using satellite data is focused. Due to their sensitivity in capturing and quantifying the complex interaction of particles and solids that contribute to turbidity levels, these particular spectral bands are preferred. The clever use of various satellites and spectral bands highlights how versatile remote sensing is for figuring out the intricate dynamics of turbidity in a variety of aquatic environments. To build connections and gather information about turbidity, these bands can be employed alone or in conjunction with other bands.

[12][16]. To analyze turbidity, the wavelengths of the green band have also been included in ratios with other bands. [7], The picture was taken using the Landsat 5 Thematic Mapper (TM). Using the regression model, turbidity is calculated.

No.	Variables	R	Std. Error of the Estimate
1	Turbidity and $B3/(B1+B2+B3)$	0.461	10.7484
2	Turbidity and B3	0.247	11.7382
3	Ln (Turbidity) and B3	0.171	

Table 1. Models with correlation coefficients [12].

Algorithms used [8] Support vector regression, random forest, K-nearest neighbor, classification and regression tree, gradient-boosted decision tree, extreme learning machine/feedforward neural network, and backpropagation. Using validation samples, the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute errors (MAE) are used to calculate the model accuracies. Models for measuring water turbidity are evaluated and chosen based on which ones produce the highest R^2 value while also displaying the lowest Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). This stringent criterion denotes the search for models that exhibit a strong correlation between predicted and observed turbidity levels (as shown by the R^2 coefficient), as well as a high level of precision and little residual error [11]. The study defined three Optical Water Types (OWTs): OWT C1, OWT C2, and OWT C3, based on the reflectance characteristics. These types of water had different characteristics in terms of pH, SPM, TP, SDD, aCDOM(), aph(), and ad(). A neural net model called BP-TURB was created for each OWT using k-mean clustering and iterative analysis [11]. The models did well at estimating turbidity, with high R^2 values and good relationships between measured and estimated turbidity.

(c) Summary and Limitations

Finding the cause of turbidity from remote sensing data can be challenging as turbidity is influenced by various factors, turbidity can have multiple causes including sedimentation, organic matter, algal blooms, pollution, and natural processes. Therefore, a comprehensive analysis combining remote sensing data, ancillary information, and field observations is crucial to identify the potential causes of turbidity accurately.

IV. DISSOLVED OXYGEN

(a) Introduction

Dissolved oxygen (DO) is the measurement of oxygen gas dissolved in water. It serves as a vital indicator of the health and quality of aquatic ecosystems. Aquatic organisms like fish and invertebrates rely on dissolved oxygen for respiration. Adequate levels of dissolved oxygen are essential for their survival and the overall balance of aquatic life. Temperature, salinity, turbulence, pollution levels, and other variables can all affect the water's dissolved oxygen content. Hypoxia, when oxygen concentrations are too low to support aquatic life, can result from low amounts of dissolved oxygen. For water bodies to remain healthy and diverse, proper dissolved oxygen levels must be monitored and maintained. Scientists and environmentalists may learn a lot about the effects of human activity and the overall health of aquatic ecosystems by measuring the quantities of dissolved oxygen in the water.

(b) Methods

By building an inversion model based on the relationship between spectral reflectance and the concentration of dissolved oxygen, one can use remote sensing data to identify dissolved oxygen [14]. The reflectance of specific satellite bands or combinations of bands is analyzed to determine the correlation coefficient with dissolved oxygen content. The band with the strongest correlation is selected for the inversion model. Water bodies are extracted by

$$MNDWI = \frac{(GREEN - SWIR1)}{(GREEN + SWIR1)}, (3)$$

SWIR- 6 band of Landsat 8 OLI, Green- 3 Band of Landsat 8 OLI with 30 m spatial resolution. Dissolved oxygen has less correlation highest is 0.5942 so a combination of these 2 bands is taken (band 1 & band 2). Dissolved oxygen (DO) is measured during in situ measurements using standard methods. Average dissolved oxygen (DO) levels, a vital indicator of water quality, are accurately measured at a respectable 11.4 mg/L. The preservation and transportation of water samples during in situ measurements were methodically carried out by the highly acclaimed ISO 5667-4, 2016 water quality sampling requirements, ensuring strict adherence to protocol. The in-situ data was meticulously processed and analyzed after that, and the result was its visual representation within a Geographic Information System (GIS) framework. This spatial depiction efficiently shows the distribution trends present in the acquired in situ measurements, giving a thorough understanding of the DO status of the nearby aquatic environment. The parameters under scrutiny were also meticulously compared against two important benchmarks, the Turkish Surface Water Quality Management Directive (SWQMD) and the directive 2006/44/EC of the European Parliament and Council, which serve as crucial standards for evaluating and validating the water quality within freshwater systems.

In the paper [13] Sentinel 2A (S2A), Landsat 8 OLI (L8 OLI), and Göktürk-2 (GK2) were the three satellite systems used. Water quality projections benefit from the moderate spatial resolution of L8 OLI and S2A pictures, whereas GK2's 2.5-m resolution gives precise small-scale dispersion. Bilinear interpolation was used to downscale L8 OLI and S2A pictures to 5 m pixels after atmospheric adjustment, to predict surface water quality, MLR, SVM, and ANN were used. The integration of numerous pictures and bands required the application of PCA to reduce data size. The PCA-based RSR approach, as well as the band-only MLR, SVM, and ANN methods, all underwent accuracy assessments. NDWI was used to assess the size of the lake, and MLR, SVM, ANN, and PCA models predicted water quality concentrations.

(c) Summary and Limitations

The limitation of dissolved oxygen detection using remote sensing data is that the correlation between dissolved oxygen content and spectral reflectance is not strong enough [14]. In the study mentioned, the correlation coefficient between dissolved oxygen and the first band is 0.5942, which indicates a moderate correlation. This suggests that the accuracy of estimating dissolved oxygen using remote sensing data may be lower compared to other water quality parameters [13]. It is crucial to remember that many factors, including air conditions, sensor calibration, and the existence of additional water elements that may change the spectral reflectance, might affect how accurately DO is estimated using remote sensing techniques. These factors can introduce uncertainties in the estimation of DO from satellite images.

V. NITROGEN AND PHOSPHORUS

(a) Introduction

Nitrogen and phosphorus are fundamental nutrients required for healthy ecosystem functioning, but their excessive presence in water bodies can lead to significant water quality issues. Elevated levels of nitrogen and phosphorus, often referred to as nutrient pollution, can result in harmful effects on water quality and aquatic ecosystems through a process called eutrophication. Excess nitrogen and phosphorus can stimulate the rapid growth of algae and aquatic plants in a water body, leading to an overabundance of plant material. As these plants die and decompose, microbial activity consumes oxygen in the water, causing a decrease in dissolved oxygen levels. This can lead to hypoxic or anoxic conditions, where oxygen concentrations become dangerously low, threatening the survival of fish and other aquatic organisms.

The precise management of water quality hinges upon a meticulous understanding of the levels of phosphorus (P) and nitrogen (N) present within aquatic ecosystems. The dynamic interplay of these essential elements significantly influences water quality and underscores their pivotal role in fostering high-quality animal husbandry practices and sustainable utilization of vital grassland resources. The imperative to monitor and accurately quantify fluctuations in N and P concentrations remains paramount for driving advancements in animal husbandry techniques and ensuring the judicious exploitation of grassland assets. Furthermore, the concentrations of nitrogen (N), phosphorus (P), and potassium (K) in alpine natural grasslands hold profound significance as robust indicators of fodder quality, deeply intertwined with optimal plant growth, development, and reproductive processes. This interdependence underscores their indispensable contribution to fortifying the overall health and vitality of ecosystems reliant on these resources [10].

(b) Methods

Amid a thorough evaluation of WorldView-2 image bands and ground truth data, an intriguing alignment becomes apparent. Within the estimation domain, the capabilities of WorldView-2's level LV3D data, encompassing geometric correction, orthorectification, and radiance calibration, emerge as pivotal in Total Phosphorus (TP) estimation. The strategic employment of Band 1 and Band 2 within this image framework seamlessly incorporates linear regression analysis, interweaving [11] into the investigative fabric. Delving deeper,

the band ratio approach unveils its significance. Within this intricate web of computations, the division of Band 8 by Band 2, a harmonious amalgamation of Near-Infrared (Nir2) and blue wavelengths, emerges as a prime candidate for TP correlation. This dynamic connection culminates in an impressive correlation coefficient, denoted as R, and stands at a compelling 0.68. This outcome underscores the effectiveness of this spectral synergy in capturing the intricate intricacies of Total Phosphorus delineation.

Worldview-2 picture bands and ground truth data are correlated. Utilizing Worldview-2 of Level LV3D, which offers geometric correction, orthorectification, and radiance correction, one can estimate TN (Total Nitrogen). After performing a linear regression analysis, Band 8 and Band 7 are utilized for TP. Ratio-banding scheme with a r value of 0.93, band 8/band 1 (Nir2/coastal blue) had the best TP correlation [17].

(c) Summary and Limitations

A possible method for precisely mapping the amounts of TP and TN in water bodies is the combination of multispectral sensors, such as WorldView-2, with sophisticated data analysis techniques, like band ratio schemes and linear regression analysis. These findings have important ramifications for maintaining and monitoring water quality, making sure that grassland resources are used sustainably, and improving high-quality animal husbandry [10]. For total phosphorus estimation, Band 8 divided by Band 2 (Nir2/blue) showed the highest correlation ($r = 0.68$) with TP levels. This band ratio scheme is the most effective in estimating TP content in water bodies [11]. For total nitrogen estimation, Band 8 divided by Band 1 (Nir2/coastal blue) demonstrated the highest correlation ($r = 0.93$) with TN levels. This band ratio scheme is highly accurate in estimating TN content in water bodies [11].

VI. TEMPERATURE AND pH

(a) Introduction

Temperature assumes a pivotal role in the thorough examination of water quality, casting a direct and extensive influence across an array of parameters. Its consequential reach extends into areas encompassing dissolved oxygen levels, the propagation of algal blooms, and more. In the aquatic realm, temperature orchestrates a symphony of intricate interactions that span physical, chemical, and biological processes within lakes and reservoirs. From governing the kinetics of reactions and rates of metabolism to shaping the dispersion of species and the dynamics of nutrients, temperature emerges as a central element governing the very essence of aquatic ecosystems [16].

Concurrently, the pH factor, denoting hydrogen ion potential, garners significant attention due to its wide-ranging impact. As underscored by the U.S. EPA (U.S. Environmental Protection Agency), pH wields a profound influence over chemical and biological reactions in water. This dynamic inter-play extends further, delineating the boundaries within which species proliferate within aquatic ecosystems. An optimal pH range of 6.5 to 8 emerges as a haven for aquatic life, fostering thriving within this pH spectrum. Venturing into freshwater domains, typical pH levels fall within the range of 6.5 to 9. The combined assessment of pH and temperature thus surfaces as an invaluable tool, facilitating vigilant surveillance of aquatic well-being and the judicious oversight of practices related to industrial wastewater discharge.

(b) Methods & Discussions

Other thermal sensors include the Thematic Mapper (TM), Enhanced Thematic Mapper Plus (ETM+) on MODIS, and Landsat-7 ETM+, cite ref21. The Thermal Infrared (TIR) band's availability aids in estimating the temperature of the water surface and land surface. The formula used to calculate the SWT, or surface water temperature, is

$$T = \frac{K2}{\ln\left(\frac{K1}{L_{\text{sensor}}} + 1\right)} \quad (4)$$

where K1 and K2 are calibration constants, with values of 666.09 W/(m² sr μm) and 1282.71 K, respectively, for Landsat 7 ETM+ and the top-of-atmosphere (TOA) radiance as measured by the sensor [16] is $L_{\text{ext,sensor}}$. By comparing in situ measurements with sensor reflectance values gathered over a lake, the author of [2] examines the failure of a model to forecast pH. They also go over a method to connect pH readings with other water quality indicators including salinity, temperature, and dissolved oxygen [16].

VII. CONCLUSION

Monitoring water quality using remote sensing and Geographic Information Systems (GIS) is a cost-effective and powerful strategy. Remote sensing data, such as satellite imagery, can be used to map water quality parameters like dissolved oxygen, turbidity, and chlorophyll. It also helps in identifying issues like algal blooms and pollution sources. Integrating GIS enables comprehensive watershed management. Real-time monitoring with remote sensors and buoys enhances responsiveness to water quality changes, contributing to the protection of water bodies and sustaining aquatic ecosystems. However, this approach has limitations. One significant challenge is the spatial resolution of remote sensing data, especially from satellites, which may not capture small-scale variations in water quality. Atmospheric conditions, such as clouds and haze, can also affect data accuracy. Moreover, the temporal coverage may not align with desired monitoring intervals, making real-time assessment difficult. Obtaining ground truth data for water quality parameters is challenging due to dynamic water bodies, logistical complexities, specialized equipment, weather, and resource limitations. Remote sensing and GIS must be validated with ground truth data to ensure accurate results. While remote sensing provides valuable insights, on-site validation, and data calibration are

VIII. REFERENCES

- [1] Topp, S.N., Pavelsky, T.M., Jensen, D., Simard, M. and Ross, M.R., 2020. Research trends in the use of remote sensing for inland water quality science: Moving towards multidisciplinary applications. *Water*, 12(1), p.169.
- [2] Hossain, A.A., Mathias, C. and Blanton, R., 2021. Remote sensing of turbidity in the Tennessee River using Landsat 8 satellite. *Remote Sensing*, 13(18), p.3785.
- [3] Iteralp, M., Ariman, S. and Aptoula, E., 2021. A deep multitask semisupervised learning approach for chlorophyll-a retrieval from remote sensing images. *Remote Sensing*, 14(1), p.18.
- [4] Yang, H., Kong, J., Hu, H., Du, Y., Gao, M. and Chen, F., 2022. A review of remote sensing for water quality retrieval: progress and challenges. *Remote Sensing*, 14(8), p.1770.
- [5] Neves, V.H., Pace, G., Delegido, J. and Antunes, S.C., 2021. Chlorophyll and suspended solids estimation in Portuguese reservoirs (Aguieira and Alqueva) from Sentinel-2 imagery. *Water*, 13(18), p.2479.
- [1] Aranha, T.R.B.T., Martinez, J.M., Souza, E.P., Barros, M.U. and Martins, E.S.P., 2022. Remote analysis of the chlorophyll-a concentration using Sentinel-2 MSI images in a semiarid environment in Northeastern Brazil. *Water*, 14(3), p.451.
- [2] Abdelal, Q., Assaf, M.N., Al-Rawabdeh, A., Arabasi, S. and Rawashdeh, N.A., 2022. Assessment of Sentinel-2 and Landsat-8 OLI for small-scale inland water quality modeling and monitoring based on handheld hyperspectral ground truthing. *Journal of Sensors*, 2022.
- [3] Ma, Y., Song, K., Wen, Z., Liu, G., Shang, Y., Lyu, L., Du, J., Yang, Q., Li, S., Tao, H. and Hou, J., 2021. Remote sensing of turbidity for lakes in northeast China using Sentinel-2 images with machine learning algorithms. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14, pp.9132-9146.
- [4] Zhao, X., Xu, H., Ding, Z., Wang, D., Deng, Z., Wang, Y., Wu, T., Li, W., Lu, Z. and Wang, G., 2021. Comparing deep learning with several typical methods in prediction of assessing chlorophyll-a by remote sensing: A case study in Taihu Lake, China. *Water Supply*, 21(7), pp.3710-3724.
- [5] Li, S., Kutser, T., Song, K., Liu, G. and Li, Y., 2023. Lake Turbidity Mapping Using an OWTs-bp Based Framework and Sentinel-2 Imagery. *Remote Sensing*, 15(10), p.2489.
- [6] Tao, J., Zhang, Z. and Yu, W., 2011, June. Monitoring Taihu water quality by using high-resolution satellite image. In 2011 19th International Conference on Geoinformatics (pp. 1-4). IEEE.
- [7] Kulkarni, A., 2011. Water quality retrieval from Landsat TM imagery. *Procedia Computer Science*, 6, pp.475-480.
- [8] Wang, J., Zhang, J., Li, T. and Wang, X., 2018, July. Water quality analysis of remote sensing images based on inversion model. In IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium (pp. 4861-4864). IEEE.
- [9] Batur, E. and Maktav, D., 2018. Assessment of surface water quality by using satellite images fusion based on PCA method in the Lake Gala, Turkey. *IEEE Transactions on Geoscience and Remote Sensing*, 57(5), pp.2983-2989.
- [10] Mohamed, M. and Plante, R., 2002, June. Remote sensing and geographic information systems (GIS) for developing countries. In IEEE International Geoscience and Remote Sensing Symposium (Vol. 4, pp. 2285-2287). IEEE.
- [11] Tavares, M.H., Cunha, A.H.F., Motta-Marques, D., Ruhoff, A.L., Cav-alcanti, J.R., Fragoso Jr, C.R., Mart'in Bravo, J., Munar, A.M., Fan, F.M. and Rodrigues, L.H.R., 2019. Comparison of methods to estimate lake-surface-water temperature using Landsat 7 ETM+ and MODIS imagery: Case study of a large shallow subtropical lake in southern Brazil. *Water*, 11(1), p.168.
- [12] Zhang, X., Liang, T., Gao, J., Zhang, D., Liu, J., Feng, Q., Wu, C. and Wang, Z., 2023. Mapping the forage nitrogen, phosphorus, and potassium contents of alpine grasslands by integrating Sentinel-2 and Tiangong-2 data. *Plant Methods*, 19(1), p.48.